

CHELFF, Algeria

WMO: 604250

Lat: 36.213N

Lon: 1.332E

Elev: 463

StdP: 14.45

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	38.8	40.8	30.4	25.0	59.6	33.1	28.1	55.5	31.6	57.3	27.9	56.3	3.5	60	0.435

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.0	105.9	71.8	102.6	71.8	100.1	71.1	77.6	95.8	75.7	93.3	74.3	91.6	9.4	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.7	123.4	85.9	70.6	114.6	83.8	69.1	108.7	83.0	41.3	95.3	39.4	94.1	38.0	91.7	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	20.2	17.6	34.2	111.6	2.2	2.2	32.6	113.2	31.3	114.5	30.1	115.8	28.5	117.4
			33.4	79.6	1.8	3.8	32.1	82.3	31.1	84.5	30.1	86.6	28.8	89.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	68.3	52.2	53.7	58.7	63.0	71.4	80.3	87.0	86.7	79.3	71.7	60.1	54.5	(6)
	DBStd	13.45	4.15	5.24	5.14	5.93	6.61	6.43	4.61	4.28	4.65	6.00	5.74	4.53	(7)
	HDD50	62	24	22	3	0	0	0	0	0	0	0	1	12	(8)
	HDD65	1550	396	316	205	107	17	0	0	0	0	13	169	327	(9)
	CDD50	6744	93	126	272	391	662	909	1146	1139	880	672	303	151	(10)
	CDD65	2757	0	1	8	48	214	460	681	674	430	220	20	1	(11)
	CDH74	33051	0	6	93	497	2182	5590	9486	9135	4261	1686	113	2	(12)
	CDH80	18275	0	1	11	138	965	3140	5886	5494	2011	610	19	0	(13)
(14) Wind	WSAvg	6.4	6.2	6.9	6.8	6.9	7.0	6.9	6.7	6.3	6.1	5.5	6.0	5.5	(14)
(15) Precipitation	PrecAvg	16.7	2.0	2.1	1.7	1.7	1.2	0.4	0.0	0.2	0.6	1.5	2.2	2.1	(15)
	PrecMax	25.2	7.0	5.6	5.0	5.4	3.9	2.0	0.6	1.6	2.9	5.2	6.8	5.1	(16)
	PrecMin	10.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	(17)
	PrecStd	4.4	1.3	1.6	1.1	1.4	1.1	0.4	0.1	0.4	0.7	1.4	1.5	1.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.6	73.7	81.0	89.5	100.6	107.0	111.4	109.4	101.8	95.0	82.5	72.0	(19)
		MCWB	57.3	57.7	60.2	67.3	70.8	70.1	72.4	73.3	72.6	67.5	64.6	58.9	(20)
	2%	DB	65.1	68.7	76.8	83.8	93.5	102.4	106.3	105.3	96.6	90.0	76.9	67.9	(21)
		MCWB	55.2	56.4	59.6	64.0	67.5	69.6	72.9	71.4	70.7	66.5	61.3	56.8	(22)
	5%	DB	62.4	66.0	72.7	79.6	89.4	98.7	103.2	102.1	93.5	86.3	73.1	64.8	(23)
		MCWB	54.1	55.5	58.1	62.4	66.1	69.7	72.2	72.3	70.2	65.7	60.3	55.7	(24)
	10%	DB	59.7	63.0	69.5	75.4	85.3	95.0	100.2	99.0	90.2	82.7	69.6	62.3	(25)
		MCWB	52.9	54.0	57.2	60.7	65.2	68.4	71.4	71.3	69.2	64.9	59.3	54.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	58.7	61.1	65.2	69.4	74.4	76.8	80.7	81.0	77.9	71.7	66.9	60.7	(27)
		MCDB	65.8	69.1	76.2	84.4	93.4	96.0	100.2	97.8	93.6	85.8	78.7	69.4	(28)
	2%	WB	56.5	58.5	61.8	65.9	70.3	73.7	76.9	78.2	74.6	69.4	63.8	58.5	(29)
		MCDB	62.9	66.1	71.6	79.9	87.6	94.3	97.4	96.1	88.1	83.3	72.3	65.6	(30)
	5%	WB	55.0	56.5	59.9	63.7	68.2	71.9	75.1	76.2	72.9	67.8	61.8	56.8	(31)
		MCDB	60.8	63.7	69.2	76.8	84.8	92.6	94.9	93.1	86.7	81.3	70.0	62.9	(32)
	10%	WB	53.5	54.8	58.1	61.6	66.4	70.2	73.7	74.6	71.5	66.6	60.1	55.3	(33)
		MCDB	58.9	61.6	67.1	73.3	82.1	89.1	93.1	90.9	84.8	79.1	67.7	60.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.3	17.2	19.2	21.1	22.6	24.8	26.0	24.7	21.8	20.1	15.7	15.1	(35)
		MCDBR	18.7	21.5	25.0	27.9	28.9	29.5	29.7	28.4	25.9	24.6	20.7	18.4	(36)
	5% WB	MCWBR	11.1	11.7	12.5	13.0	11.5	9.9	10.5	9.8	9.7	9.2	10.0	10.3	(37)
		MCWBR	16.4	18.7	21.3	25.3	25.9	26.7	26.7	24.4	22.3	21.1	18.3	16.2	(38)
(40) Clear-Sky Solar Irradiance	taub		0.325	0.346	0.404	0.424	0.456	0.469	0.518	0.488	0.457	0.408	0.367	0.332	(40)
		taud	2.489	2.424	2.237	2.203	2.136	2.114	1.971	2.062	2.163	2.296	2.391	2.484	(41)
	Ebn at Noon		276	282	273	273	265	260	246	252	253	256	256	265	(42)
		Edh at Noon	26	31	41	45	49	50	57	51	44	36	29	25	(43)
(44) All-Sky Solar Radiation	RadAvg		809	1064	1466	1796	2079	2372	2336	2093	1641	1250	856	723	(44)
	RadStd		83	117	110	140	160	88	50	85	79	90	73	51	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	+0.74	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page